

Supplementary Materials for Paper Extreme Value Theory-Driven Robust Feature Selection with Application to Estimation of Interaction Stiffness

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S1. STATISTICAL RESULTS

Tables [S1](#) – [S4](#) provide the statistical results corresponding to Section IV-C, reporting the classification accuracies with standard deviations, where the best results are highlighted in bold. The “Win” and “Loss” indicators denote the number of datasets on which NF-MRMR outperforms or underperforms the competing methods, respectively. The “Rank” represents the average ranking based on classification accuracies across

the 15 datasets.

S2. EXTRACTED SEMG FEATURES

Table [S5](#) presents the top 10 features selected by NF-MRMR in Section V.

Table [S6](#) presents the categories of the sEMG signals extracted in Section V.

TABLE S1
COMPARISON OF CLASSIFICATION ACCURACY (MEAN±STD %) FOR TEN FEATURE SELECTION METHODS USING DT CLASSIFIER.

Dataset	mRMR	MNFR-MR	GradEnFS	MRMSR	ANMVFS	C-FSAE	WAST	TAGA	BMFS	InfFS	CNAFS	NF-MRMR
Chia	51.62±0.82	77.57±0.41	58.30±0.77	73.68±0.48	51.54±0.56	67.30±0.49	42.01±0.69	74.61±0.54	44.50±0.76	64.16±0.63	59.69±0.68	78.82±0.60
Chri	86.00±0.30	99.56±0.04	94.86±0.23	96.73±0.18	94.86±0.15	97.86±0.13	86.17±0.31	99.77±0.05	93.10±0.22	97.04±0.07	94.06±0.21	99.53±0.05
Coil	56.25±0.10	61.70±0.17	53.26±0.26	53.46±0.15	52.36±0.18	18.57±0.06	49.27±0.14	58.60±0.12	50.46±0.19	51.02±0.20	57.87±0.22	59.73±0.18
Fash	53.54±0.03	58.15±0.05	51.82±0.05	36.37±0.06	49.98±0.04	12.68±0.02	54.11±0.05	51.11±0.04	37.33±0.03	45.62±0.05	18.16±0.02	61.05±0.05
Glio	63.23±0.72	61.67±0.77	57.35±1.12	68.64±0.73	45.98±0.96	45.80±0.36	58.05±0.90	54.77±1.02	40.16±0.93	41.75±0.64	40.13±0.83	67.60±0.75
Lung	78.39±0.29	88.04±0.44	74.78±0.45	83.60±0.34	76.36±0.27	79.23±0.39	67.79±0.43	80.06±0.32	79.24±0.36	79.87±0.40	71.75±0.49	88.53±0.42
Orl	30.53±0.23	29.62±0.35	29.35±0.21	31.33±0.21	25.39±0.19	20.78±0.21	19.10±0.29	24.86±0.17	17.84±0.26	18.63±0.34	23.40±0.22	29.46±0.24
Orlr	71.52±0.43	73.02±0.33	63.83±0.54	73.00±0.78	57.98±0.85	48.60±0.60	50.44±0.69	74.01±0.59	55.98±0.44	48.57±0.40	56.60±0.41	77.45±0.52
Pixr	95.76±0.10	95.42±0.19	84.25±0.44	91.11±0.15	91.88±0.17	77.82±0.45	91.95±0.26	95.56±0.15	80.82±0.42	72.76±0.39	75.64±0.57	97.19±0.17
Smkc	56.49±0.75	66.90±0.80	58.89±0.61	60.81±0.68	54.66±0.36	62.26±0.39	59.68±0.67	65.04±0.65	66.51±0.56	55.83±0.78	54.76±0.69	66.57±0.57
Subr	55.26±0.91	80.10±1.14	57.36±0.86	65.76±1.11	54.24±0.67	64.83±0.77	52.46±1.24	82.16±0.94	79.92±0.72	50.20±0.95	54.88±1.06	77.86±0.89
Waar	55.36±0.51	51.15±0.63	53.02±0.51	61.83±0.53	34.56±0.65	33.40±0.61	45.02±0.69	64.04±0.53	38.97±0.54	33.21±0.50	24.58±0.59	67.83±0.63
Wpie	64.72±0.58	68.39±0.51	61.21±0.32	70.67±0.46	57.55±0.42	50.02±0.46	40.74±0.32	62.73±0.45	60.27±0.36	55.95±0.39	46.21±0.38	70.98±0.29
Yale	40.32±0.56	39.24±0.44	41.97±0.45	46.03±0.55	38.69±0.47	31.50±0.48	27.57±0.31	48.29±0.58	21.23±0.57	25.41±0.58	35.41±0.45	49.19±0.47
Yeoh	35.45±0.43	83.97±0.24	61.63±0.40	81.22±0.34	38.71±0.33	74.57±0.35	42.62±0.49	40.82±0.44	64.61±0.35	64.70±0.38	71.78±0.26	84.72±0.25
Accuracy	59.63±0.45	68.97±0.43	60.13±0.48	66.28±0.45	54.98±0.42	52.35±0.38	52.47±0.5	65.10±0.44	55.40±0.45	53.65±0.45	52.33±0.47	71.77±0.41
Win/Loss	15 / 0	13 / 2	15 / 0	13 / 2	15 / 0	15 / 0	15 / 0	13 / 2	15 / 0	15 / 0	15 / 0	-
Rank	6.13	2.87	6.67	4.27	8.33	8.20	9.00	4.00	8.47	9.13	9.13	1.73

TABLE S2
COMPARISON OF CLASSIFICATION ACCURACY (MEAN±STD %) FOR TEN FEATURE SELECTION METHODS USING KNN CLASSIFIER.

Dataset	mRMR	MNFR-MR	GradEnFS	MRMSR	ANMVFS	C-FSAE	WAST	TAGA	BMFS	InfFS	CNAFS	NF-MRMR
Chia	57.45±0.42	87.04±0.28	68.63±0.36	74.42±0.35	55.94±0.32	69.98±0.21	54.07±0.39	81.37±0.24	51.55±0.39	71.32±0.49	66.84±0.31	85.82±0.31
Chri	84.66±0.19	99.64±0.03	95.91±0.12	98.69±0.05	94.51±0.17	98.06±0.05	88.16±0.18	99.80±0.01	96.55±0.11	96.28±0.06	94.69±0.12	99.65±0.03
Coil	79.36±0.09	81.07±0.07	80.20±0.05	69.12±0.13	70.77±0.04	17.42±0.05	60.61±0.06	80.53±0.07	64.65±0.12	65.24±0.06	75.48±0.09	84.19±0.09
Fash	55.42±0.06	64.47±0.04	58.00±0.03	43.20±0.07	56.20±0.09	13.30±0.02	60.69±0.06	56.28±0.04	40.65±0.03	52.13±0.04	19.98±0.05	68.49±0.04
Glio	54.18±0.63	75.51±0.62	66.85±0.72	62.32±0.56	51.40±0.76	37.07±0.48	61.01±1.01	67.12±0.59	33.56±0.85	40.74±0.83	47.16±0.59	80.26±0.61
Lung	82.87±0.15	92.59±0.07	83.33±0.13	88.55±0.12	80.92±0.17	83.82±0.18	78.55±0.19	82.04±0.13	84.56±0.17	84.77±0.23	76.99±0.15	94.77±0.09
Orl	43.50±0.21	42.72±0.28	47.71±0.27	49.27±0.19	45.27±0.24	31.98±0.24	20.08±0.20	43.13±0.18	18.37±0.16	22.97±0.19	44.70±0.22	58.92±0.21
Orlr	58.11±0.44	57.96±0.39	64.92±0.50	72.30±0.35	47.78±0.44	54.06±0.40	42.16±0.47	78.21±0.54	45.92±0.37	39.97±0.61	46.61±0.44	79.90±0.25
Pixr	81.04±0.26	81.32±0.24	71.03±0.19	72.26±0.15	83.20±0.16	63.72±0.29	67.90±0.21	93.70±0.09	76.96±0.24	48.53±0.35	57.57±0.35	95.26±0.11
Smkc	61.00±0.53	74.99±0.35	69.13±0.47	68.02±0.46	60.36±0.50	68.60±0.31	63.96±0.33	73.41±0.27	69.70±0.25	60.07±0.54	56.56±0.44	76.69±0.23
Subr	67.14±0.59	81.48±0.38	64.66±0.45	73.99±0.64	64.34±0.51	66.61±0.49	64.71±0.43	89.46±0.33	91.15±0.30	64.73±0.53	65.00±0.56	81.70±0.43
Waar	48.39±0.32	45.24±0.32	44.58±0.37	55.75±0.34	24.19±0.38	31.70±0.38	39.77±0.46	61.04±0.45	36.39±0.44	28.99±0.43	19.55±0.32	66.77±0.36
Wpie	66.74±0.41	72.43±0.23	66.34±0.28	77.70±0.26	63.13±0.32	43.12±0.36	29.98±0.48	64.41±0.41	62.60±0.36	37.16±0.30	52.57±0.50	81.55±0.34
Yale	35.35±0.35	35.58±0.38	34.08±0.24	48.39±0.28	39.16±0.33	24.07±0.26	21.99±0.27	48.48±0.36	24.56±0.46	24.69±0.42	28.35±0.23	56.16±0.35
Yeoh	37.38±0.33	90.27±0.15	69.85±0.20	88.84±0.12	44.04±0.33	79.97±0.22	47.42±0.33	42.19±0.36	72.98±0.26	65.86±0.27	78.14±0.17	91.36±0.10
Accuracy	60.84±0.33	72.15±0.26	65.68±0.29	69.52±0.27	58.75±0.32	52.23±0.26	53.4±0.34	70.74±0.27	58.01±0.30	53.56±0.36	55.35±0.30	80.10±0.24
Win/Loss	15 / 0	14 / 1	15 / 0	15 / 0	15 / 0	15 / 0	15 / 0	14 / 1	14 / 1	15 / 0	15 / 0	-
Rank	7.00	3.40	6.00	4.53	8.00	8.33	9.47	4.07	8.00	9.00	8.93	1.27

TABLE S3
COMPARISON OF CLASSIFICATION ACCURACY (MEAN±STD %) FOR TEN FEATURE SELECTION METHODS USING NN CLASSIFIER.

Dataset	mRMR	MNFR-MR	GradEnFS	MRMSR	ANMVFS	C-FSAE	WAST	TAGA	BMFS	InfFS	CNAFS	NF-MRMR
Chia	51.38±0.76	83.45±0.45	63.42±0.64	73.34±0.57	48.18±0.71	69.19±0.61	42.21±0.76	78.32±0.44	41.91±0.64	67.65±0.66	62.91±0.47	82.28±0.47
Chri	86.58±0.21	99.61±0.04	96.48±0.15	99.13±0.09	95.27±0.17	98.51±0.09	89.49±0.30	99.84±0.04	96.28±0.10	97.24±0.13	95.99±0.20	99.58±0.04
Coil	81.10±0.20	82.57±0.20	82.61±0.15	75.13±0.26	71.59±0.06	18.13±0.07	62.43±0.19	80.42±0.19	67.49±0.25	65.85±0.14	77.89±0.19	83.55±0.20
Fash	58.53±0.04	64.63±0.07	59.73±0.04	42.44±0.07	57.33±0.20	13.34±0.01	62.14±0.06	58.09±0.06	39.04±0.07	52.72±0.07	20.85±0.02	69.05±0.06
Glio	52.83±0.99	69.64±0.61	62.68±0.68	62.74±1.18	50.96±1.01	46.73±0.90	60.38±0.85	63.36±0.89	39.02±1.16	36.65±1.18	43.82±0.80	83.06±0.97
Lung	78.58±0.32	91.29±0.23	79.67±0.44	87.66±0.29	78.21±0.39	81.06±0.47	72.74±0.48	78.29±0.29	82.54±0.36	82.51±0.21	72.90±0.41	92.29±0.24
Orl	45.69±0.38	45.29±0.39	41.03±0.28	46.99±0.42	38.56±0.42	28.10±0.33	21.18±0.53	37.95±0.36	19.28±0.35	25.37±0.25	38.52±0.43	50.26±0.53
Orlr	69.88±0.88	70.74±0.79	68.93±0.58	78.27±0.50	59.59±0.58	52.54±0.96	54.29±0.80	79.67±0.52	58.42±0.95	50.43±0.66	50.96±0.63	86.26±0.48
Pixr	91.92±0.45	92.41±0.50	85.97±0.41	88.00±0.42	88.52±0.61	68.82±0.68	87.15±0.62	92.43±0.67	79.06±0.60	66.21±0.69	71.16±0.61	94.00±0.50
Smkc	57.72±0.50	72.40±0.39	66.37±0.49	61.78±0.50	57.36±0.77	66.17±0.55	62.00±0.68	65.72±0.54	65.83±0.44	56.36±0.56	56.03±0.53	72.53±0.40
Subr	60.86±1.12	84.04±0.52	59.38±1.16	71.23±1.02	57.68±0.90	59.83±0.96	53.76±0.99	83.66±0.75	86.68±0.62	56.69±0.93	52.63±1.10	84.41±0.83
Waar	53.79±0.57	52.29±0.59	61.08±0.69	66.48±0.75	38.85±0.54	35.25±0.70	52.65±0.52	66.81±0.54	39.66±0.44	36.70±0.71	26.41±0.61	70.77±0.73
Wpie	77.34±0.63	78.05±0.40	79.22±0.43	79.07±0.46	76.54±0.45	57.75±0.57	53.09±0.52	78.32±0.54	70.76±0.48	56.53±0.23	68.73±0.39	83.70±0.43
Yale	33.89±0.41	32.87±0.53	34.04±0.60	43.39±0.76	35.24±0.40	24.34±0.29	23.68±0.49	47.96±0.29	18.68±0.33	19.68±0.36	29.35±0.66	51.00±0.59
Yeoh	34.16±0.54	89.20±0.23	68.80±0.36	87.33±0.20	40.04±0.28	79.17±0.29	53.49±0.43	38.72±0.42	69.21±0.29	66.22±0.39	77.68±0.29	90.87±0.25
Accuracy	62.28±0.53	73.9±0.40	67.29±0.47	70.87±0.50	59.59±0.50	53.26±0.50	56.71±0.54	69.97±0.44	58.26±0.47	55.79±0.48	56.39±0.49	79.57±0.45
Win/Loss	15 / 0	14 / 1	15 / 0	15 / 0	15 / 0	15 / 0	15 / 0	14 / 1	14 / 1	15 / 0	15 / 0	-
Rank	6.73	3.27	5.33	4.47	8.00	8.27	9.07	4.53	8.20	9.47	9.40	1.27

TABLE S4
COMPARISON OF CLASSIFICATION ACCURACY (MEAN $\pm$ STD %) FOR TEN FEATURE SELECTION METHODS USING SVM CLASSIFIER.

Dataset	mRMR	MNFR-MR	GradEnFS	MRMSR	ANMVFS	C-FSAE	WAST	TAGA	BMFS	InfFS	CNAFS	NF-MRMR
Chia	56.10 $\pm$ 0.60	86.15$\pm$0.41	66.61 $\pm$ 0.33	75.34 $\pm$ 0.42	51.50 $\pm$ 0.50	71.77 $\pm$ 0.48	43.50 $\pm$ 0.50	77.48 $\pm$ 0.49	44.28 $\pm$ 0.57	69.98 $\pm$ 0.32	67.76 $\pm$ 0.47	85.20 $\pm$ 0.38
Chri	87.47 $\pm$ 0.16	99.39 $\pm$ 0.13	96.00 $\pm$ 0.22	99.24 $\pm$ 0.03	94.93 $\pm$ 0.15	98.84 $\pm$ 0.06	90.25 $\pm$ 0.21	99.47$\pm$0.17	96.45 $\pm$ 0.21	96.38 $\pm$ 0.14	96.22 $\pm$ 0.14	99.43 $\pm$ 0.14
Coil	85.19 $\pm$ 0.14	89.76 $\pm$ 0.11	89.85 $\pm$ 0.05	85.34 $\pm$ 0.10	81.93 $\pm$ 0.18	15.86 $\pm$ 0.10	63.63 $\pm$ 0.09	88.17 $\pm$ 0.11	77.09 $\pm$ 0.07	67.56 $\pm$ 0.16	85.22 $\pm$ 0.14	91.16$\pm$0.04
Fash	44.44 $\pm$ 0.23	61.10 $\pm$ 0.20	54.64 $\pm$ 0.22	38.86 $\pm$ 0.25	54.29 $\pm$ 0.17	10.82 $\pm$ 0.10	59.25 $\pm$ 0.09	55.06 $\pm$ 0.20	25.38 $\pm$ 0.32	48.33 $\pm$ 0.12	10.73 $\pm$ 0.11	66.64$\pm$0.15
Glio	54.33 $\pm$ 0.69	71.76 $\pm$ 0.75	65.28 $\pm$ 0.87	66.13 $\pm$ 0.95	54.32 $\pm$ 0.81	43.61 $\pm$ 0.50	59.60 $\pm$ 0.85	66.22 $\pm$ 0.75	40.05 $\pm$ 0.66	38.92 $\pm$ 0.60	49.07 $\pm$ 1.00	84.62$\pm$0.57
Lung	72.82 $\pm$ 0.69	92.31 $\pm$ 0.35	83.31 $\pm$ 0.24	90.02 $\pm$ 0.26	81.01 $\pm$ 0.62	83.30 $\pm$ 0.27	79.48 $\pm$ 0.31	82.82 $\pm$ 0.33	84.82 $\pm$ 0.35	86.02 $\pm$ 0.26	79.34 $\pm$ 0.25	93.27$\pm$0.24
Orl	68.08 $\pm$ 0.27	66.89 $\pm$ 0.20	66.59 $\pm$ 0.21	66.69 $\pm$ 0.22	60.21 $\pm$ 0.28	43.88 $\pm$ 0.21	34.00 $\pm$ 0.17	60.71 $\pm$ 0.22	31.09 $\pm$ 0.27	41.08 $\pm$ 0.28	63.56 $\pm$ 0.22	73.45$\pm$0.15
Orlr	76.67 $\pm$ 0.42	77.02 $\pm$ 0.35	83.92 $\pm$ 0.37	88.31 $\pm$ 0.21	72.43 $\pm$ 0.43	65.79 $\pm$ 0.40	63.35 $\pm$ 0.41	90.66 $\pm$ 0.28	67.46 $\pm$ 0.29	55.15 $\pm$ 0.65	61.83 $\pm$ 0.55	93.47$\pm$0.25
Pixr	96.54 $\pm$ 0.17	96.39 $\pm$ 0.10	93.34 $\pm$ 0.17	91.97 $\pm$ 0.24	90.84 $\pm$ 0.16	77.95 $\pm$ 0.48	95.04 $\pm$ 0.16	95.68 $\pm$ 0.14	82.11 $\pm$ 0.34	72.35 $\pm$ 0.46	84.51 $\pm$ 0.32	97.25$\pm$0.18
Smkc	58.44 $\pm$ 0.34	73.55 $\pm$ 0.39	69.44 $\pm$ 0.40	64.54 $\pm$ 0.58	58.22 $\pm$ 0.47	63.93 $\pm$ 0.43	59.90 $\pm$ 0.43	64.13 $\pm$ 0.57	65.67 $\pm$ 0.43	56.87 $\pm$ 0.42	55.10 $\pm$ 0.49	74.01$\pm$0.51
Subr	61.60 $\pm$ 0.95	87.93$\pm$0.42	58.28 $\pm$ 0.94	76.34 $\pm$ 0.74	59.14 $\pm$ 0.61	63.62 $\pm$ 0.77	56.47 $\pm$ 0.81	85.86 $\pm$ 0.55	88.34 $\pm$ 0.43	58.97 $\pm$ 0.73	56.72 $\pm$ 0.95	87.29 $\pm$ 0.60
Waar	61.46 $\pm$ 0.29	60.74 $\pm$ 0.57	71.92 $\pm$ 0.34	70.78 $\pm$ 0.35	46.88 $\pm$ 0.45	41.82 $\pm$ 0.50	57.81 $\pm$ 0.72	75.08 $\pm$ 0.42	48.32 $\pm$ 0.32	39.21 $\pm$ 0.53	32.82 $\pm$ 0.45	81.64$\pm$0.38
Wpie	84.63 $\pm$ 0.22	84.36 $\pm$ 0.20	87.06 $\pm$ 0.27	86.30 $\pm$ 0.15	75.22 $\pm$ 0.30	66.58 $\pm$ 0.33	56.12 $\pm$ 0.35	86.80 $\pm$ 0.19	78.72 $\pm$ 0.29	66.00 $\pm$ 0.49	79.01 $\pm$ 0.42	90.18$\pm$0.18
Yale	44.69 $\pm$ 0.40	44.45 $\pm$ 0.37	48.13 $\pm$ 0.42	52.29 $\pm$ 0.36	45.75 $\pm$ 0.46	34.60 $\pm$ 0.42	31.65 $\pm$ 0.37	55.89 $\pm$ 0.38	26.19 $\pm$ 0.38	31.04 $\pm$ 0.29	42.53 $\pm$ 0.25	62.03$\pm$0.27
Yeoh	38.89 $\pm$ 0.26	91.09 $\pm$ 0.14	71.59 $\pm$ 0.38	89.87 $\pm$ 0.16	45.58 $\pm$ 0.43	82.42 $\pm$ 0.30	54.29 $\pm$ 0.32	43.39 $\pm$ 0.41	72.28 $\pm$ 0.26	70.28 $\pm$ 0.19	82.25 $\pm$ 0.29	92.22$\pm$0.17
Accuracy	66.09 $\pm$ 0.39	78.86 $\pm$ 0.31	73.73 $\pm$ 0.36	76.13 $\pm$ 0.33	64.82 $\pm$ 0.40	57.65 $\pm$ 0.36	60.29 $\pm$ 0.39	75.16 $\pm$ 0.35	61.88 $\pm$ 0.35	59.88 $\pm$ 0.38	63.11 $\pm$ 0.40	84.79$\pm$0.28
Win/Loss	15 / 0	13 / 2	15 / 0	15 / 0	15 / 0	15 / 0	15 / 0	14 / 1	15 / 0	15 / 0	15 / 0	-
Rank	7.27	3.27	5.27	4.47	8.33	8.33	9.13	4.27	8.07	9.40	8.93	1.27

TABLE S5
TOP 10 FEATURES SELECTED BY NF-MRMR.

Rank	Feature Name	Abbreviation
1	Willison Amplitude of 3 rd sensor	WAMP-3
2	Median Frequency of 1 st sensor	MDF-1
3	Hjorth Complexity of 5 th sensor	HjorthComp-5
4	Median Frequency of 3 rd sensor	MDF-3
5	Willison Amplitude of 5 th sensor	WAMP-5
6	Spectral Entropy of 4 th sensor	SpecEntropy-4
7	Peak Frequency of 4 th sensor	PF-4
8	Willison Amplitude of 4 th sensor	WAMP-4
9	Median Frequency of 5 th sensor	MDF-5
10	Hjorth Mobility of 2 nd sensor	HjorthMob-2

TABLE S6
EXTRACTED SEMG FEATURES.

	Category	Abbreviation	Full Name
1	Time-Domain	MAV	Mean Absolute Value
2		WL	Waveform Length
3		SSI	Simple Square Integral
4		AAC	Average Amplitude Change
5		DASDV	Difference Absolute Standard Deviation Value
6		L1norm	L1 Norm
7		ZC	Zero Crossing
8		VAR	Variance
9		Energy	Signal Energy
10		WAMP	Willison Amplitude
11		SSC	Slope Sign Change
12		STD	Standard Deviation
13		RMS	Root Mean Square
14		IEMG	Integrated EMG
15		Kurtosis	Kurtosis
16		Skewness	Skewness
17		Mean	Mean Value
18		MAX	Maximum Value
19		MIN	Minimum Value
20		Range	Range
21		LOG	Logarithmic Mean Absolute Value
22		DAMV	Difference Absolute Mean Value
23		ARV	Average Rectified Value
24		WLR	Waveform Length Ratio
25	Frequency-Domain	MF	Mean Frequency
26		MDF	Median Frequency
27		PF	Peak Frequency
28		PSD_Energy	Power Spectral Density Energy
29		FreqVar	Frequency Variance
30		SpecSkew	Spectral Skewness
31		SpecEntropy	Spectral Entropy
32	Hjorth Parameters	HjorthAct	Hjorth Activity
33		HjorthMob	Hjorth Mobility
34		HjorthComp	Hjorth Complexity